CS5-GENICULAR ARTERY EMBOLISATION TO TREAT PAIN FROM KNEE OSTEOARTHRITIS(KOA)

WHAT IS GENICULAR ARTERY EMBOLIZATION (GAE) AND HOW IT CAN BE USED TO TREAT PAIN IN KNEE OSTEOARTHRITIS?

- Genicular Artery Embolisation (GAE) is a minimally invasive alternative to surgery for people who are not ready or suitable for knee replacement,
- Performed by interventional radiologists, it requires precise navigation and imaging for success
- Relatively new treatment used to manage chronic knee pain caused by knee osteoarthritis (KOA).
- It targets arteries, that supply abnormal neo vessels within the joint space, introducing particulate embolic agents to cut off the blood and nerve supply to inflamed areas, significantly reducing knee pain.

Pathology of KOA:

Osteoarthritis causes **mechanical damage in the knee/joint**. → This damage leads **to inflammation, releasing pro-inflammatory factors**. → These stimulate **new blood vessel (angiogenesis)** and **new nerve (neoinnervation) growth** in the joint space. → The region becomes **hypervascularized**, and the **new sensory nerves** begin transmitting **pain signals** during movement.

GAE Mechanism:

1. A small incision is made in the femoral artery.
2. Using image guidance (e.g., X-ray with contrast agent), an interventional radiologist navigates a guidewire and catheter into the genicular arteries (blood vessels around the knee).
3. A contrast agent highlights the hypervascular regions (pain-generating sites).-shows as blushed sites on x-ray.
4. A microcatheter is placed at the target site.
5. Embolic agents (tiny microspheres) are injected to block these abnormal blood vessels.
6. This causes ischemia (loss of blood supply), kills the vessels and associated nerves, and interrupts pain transmission.

Simple concept: Kill the blood vessels → Kill the nerves → Stop the pain.

WHAT ARE THE KEY REQUIREMENTS FOR CHOOSING THE MATERIALS FOR THE EMBOLIC AGENT SUITABLE FOR GAE?

- Particulate embolic agents are favoured
- Particle size is about 200-300 microns(has to be deliverable through microcatheter without blocking it)
- Isobouyancy in contrast agent mixtures is desirable possible (**easily suspended** with **no** or **slow sedimentation to ensure smooth delivery**)
- Radiopacity desirable (but not essential) **visibility in suspension** to aid in delivery and on Fluoroscope/CT.
- Be a sterile, off-the-shelf product with a reasonable shelf-life (3 years)-so readily available.
- Block the target site
- Should not migrate to a non-target site
- Degradable material is preferred for added safety (as per IMP/CS)
- Rate of degradation (1-3 hours) and not dependent solely on slow hydrolytic processes.
- Biocompatible / non-toxic when they break down(preferably metabolised or excreted)

WHAT IS THE BEST MATERIAL FOR THIS APPLICATION AND WHY?

SakuraBead® Temporary Embolic for GAE

Material Selection: Alginate sourced from seaweed,with embedded alginate lyase enzyme.-sustainable and biocompatible hydrogel.

1. **High Biocompatibility**-When highly purified (removal of proteins), alginate is **well-tolerated in the human body**, avoiding immune response or toxicity.

2. **Sustainable and Readily Available**-Sourced from **seaweed**, making it a **renewable, cost-effective** material for biomedical applications.

3. **Easily Processed into Microspheres**

- **Calcium crosslinking** enables simple formation of stable **hydrogel microspheres**.
- Multiple fabrication techniques are compatible: **electrostatic droplet generation, jet cutting, microfluidics**, etc.

4. Mechanical Properties Optimised for Delivery

- **Compressibility and elastic recovery** allow the microspheres to:
- Pass through microcatheters without fragmenting
- **Re-expand and remain stable** in the target arteries

5. Controlled Biodegradability

- **Alginate alone does not degrade in the body**.
- Embedding **alginate lyase enzyme** enables:
- **Controlled degradation rate**
- Consistent breakdown across patients (not dependent on individual enzymatic function).
- Degradation into fragments small enough to be **excreted via kidneys within hours**

SOME AGENTS ARE DYED OR RADIOPAQUE MICROSPHERES ARE ADDED. WHY? WHAT ARE THE ADVANTAGES AND ISSUES OF THAT?

	Dyed Microspheres	Radiopaque Microspheres
Advantages	• Easily visible in syringe • Helps confirm suspension • Mixes well with contrast • Aids smooth injection	• Can be seen on X-ray after accumulation • Useful in tumour embolisation where delivery tracking is critical
Issues / Disadvantages	• Not visible under X-ray → can't track during delivery • In degradable products, dye must be proven safe and non-toxic	• Settle quickly due to high density • Hard to suspend and inject evenly • Indistinguishable from contrast agent during delivery • Only visible once accumulated • Added density makes them less practical for GAE-when trying to deliver could be sedimenting due to added density.

HOW WOULD YOU MAKE THE EMBOLIC AGENT?

SUMMARY:

1.Mixing

- **Alginate** is mixed with **alginate lyase enzyme**

2. Microsphere Formation

- The mixture is passed through an **electrostatic encapsulator**,
- Droplets are **dripped into a calcium bath**
- **Calcium crosslinks** the alginate, forming **microspheres**

3. Enzyme Inactivation During Formation

- Process done at **low pH** to keep the enzyme **inactive**
- Prevents it from degrading the microsphere during formation

4. Freeze-Drying (Lyophilisation)

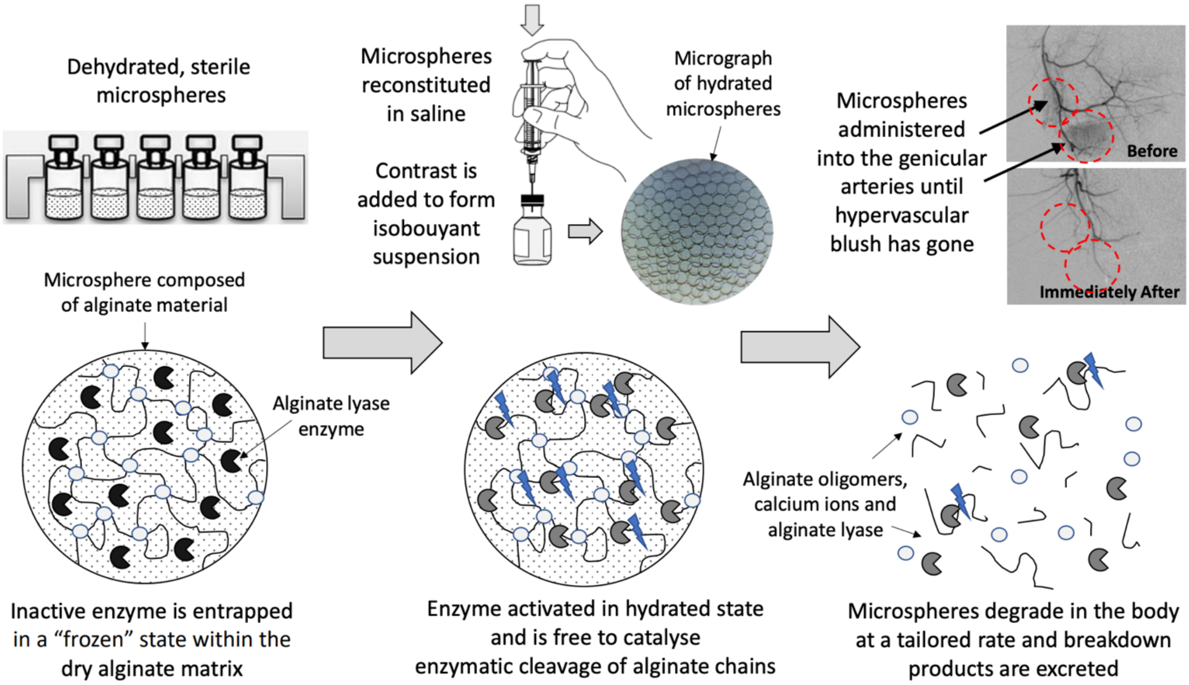
- Microspheres are placed into a **freeze-drier**
- **Water is removed**, leaving a **dry, stable powder**
- Enzyme is **“frozen” in place**, inactive without water

5. Sterilisation

- Product is **gamma irradiated** for sterilisation
- Stable for **long-term dry storage**

6.Activation: Rehydrated in saline just before use:enzyme becomes active.initiating timed degradation.

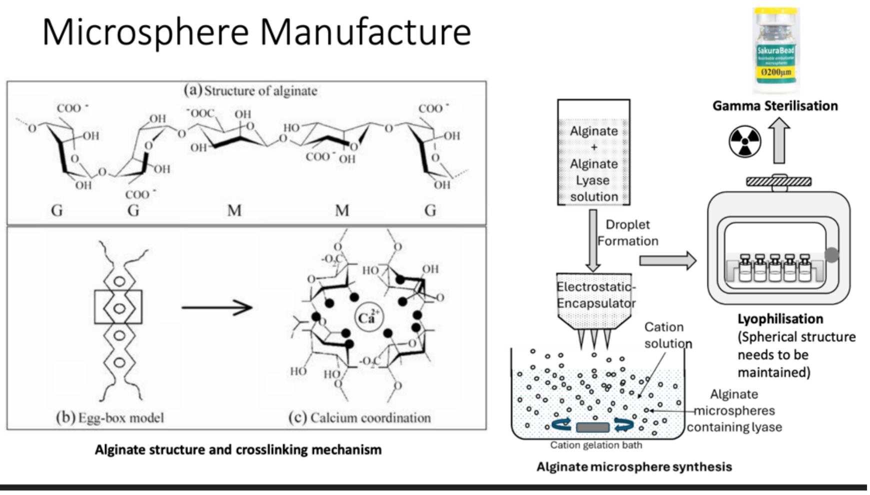
SakuraBead®: Preparation for Use



WHY DOES THE SIZE OF THE MICROSPHERE MATTER AND HOW MICROSPHERE ARE MANUFACTURED?

1. **Targeted Vessel Occlusion**:
    - Microsphere size (typically **200–300 microns** for GAE) must **match the diameter of the target blood vessels** to effectively block them and stop blood flow.
    - Too small: may **migrate** to non-target areas, causing unintended damage.
    - Too large: may **fail to reach** the targeted vessel due to size constraints in delivery.
2. **Controlled Delivery**:
    - Proper sizing ensures **predictable flow behaviour** through microcatheters.
    - Maintains **safe and uniform embolization** without catheter clogging or uneven distribution.
3. **Safety and Efficacy**:
    - Optimal size ensures **minimal systemic risk** and **maximizes local therapeutic effect** by ensuring beads lodge precisely at the treatment site.

Microsphere Manufacture



1. Alginate & Enzyme Preparation

Alginate (a polysaccharide from seaweed) is mixed with **alginate lyase enzyme**.

This mixture forms the base solution for droplet formation.

2. Droplet Formation

2. The alginate–enzyme solution is converted into droplets using one of several techniques:
    - **Electrostatic droplet generator** – uses electric charge to form uniform droplets

    - **Jet cutter** – mechanically chops a fluid stream into droplets
    - **Microfluidics platforms** – uses precise flow control in tiny channels for high uniformity
  - **IN IMAGE**: The solution is passed through an **electrostatic encapsulator** to create **uniform droplets**
  - these droplets are then **dropped into a cation-calcium (Ca^{2+}) bath**, where crosslinking occurs.

3. Calcium Crosslinking – "Egg-Box" Model
- **Calcium ions** coordinate with guluronic acid regions of alginate, forming a **gel network**.
- This stabilises the droplets into **hydrogel microspheres**.
- Shown chemically in:
- **(b) Egg-box model** and
- **(c) Calcium coordination diagram**

4. Lyophilisation (Freeze-Drying)
- The hydrogel microspheres are **freeze-dried** to:
- Remove water
- Stabilise the structure
- **Preserve the enzyme in an inactive "frozen" state**
- This process ensures **long shelf life** and a **dry powder form** that reactivates upon rehydration.

5. Gamma Sterilisation
- Final sterilisation step using **gamma radiation**
- Ensures sterility without compromising the integrity of the microsphere

End Product:
Dehydrated, sterile alginate microspheres (~200 µm in size)
Contain **inactive alginate lyase**
Ready for **rehydration and clinical use** (e.g., SakuraBead® for GAE)

EXPLAIN THE COMPRESSIBILITY AND ISOBOUYANCY PROPERTY OF THE MATERIAL AND ITS IMPORTANCE IN THIS APPLICATION.

.

Property	Definition	Importance in GAE
Compressibility	The ability of the microsphere material to deform under pressure and return to its original shape once the pressure is removed (elastic recovery).	• Allows beads to safely pass through **microcatheters** and **narrow vessels** without breaking. • Prevents **cracking or fragmentation** during injection. • Ensures **structural integrity** and **stable embolization** as beads re-expand at the target site.
Isobuoyancy	The property by which microspheres remain neutrally buoyant in contrast agent or saline—neither sinking nor floating.	• Maintains **even suspension** during preparation and injection. • Prevents **sedimentation or layering**, unlike heavier radiopaque beads. • Supports **consistent dose delivery**, **accurate embolization**, and **ease of handling** without constant remixing.